

The Structure of Coherent Construction: A Bivalent Trajectory Theory

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Abstract

We develop a bivalent trajectory theory of coherent construction in the finite fixed-scope setting. As the trajectory-theoretic member of a layered foundational program, the paper does not re-derive the standing/admissibility substrate from below; it records the fixed substrate needed for the trajectory theorems and then isolates the exact path geometry forced once that substrate is in place. Within that stated regime, Section 2 proves the local closure result needed for the trajectory layer: any admissibility-preserving auxiliary classifier on the same substrate collapses to tensor-determined bookkeeping, so no stricter standing-bearing layer is available there. Over that substrate, a trajectory is coherent if and only if every state along it retains standing. We prove closure under prefix extraction and legal composition, invariance under identity-preserving transformation, tensor determination, skin neutrality, and irrecoverability of failure under extension. The central result is a no-deferred-coherence theorem: once an incoherent prefix appears, no admissible continuation restores coherence. The theory is internal to a single fixed substrate and does not address cross-scope transport, branching developments, infinite developments, or alternative regimes with different scope data. Within that stated regime, coherence emerges as a binary structural invariant of admissible construction histories with a uniquely determined finite fixed-scope path structure.

Contents

1	Introduction	3
2	Formal setup	4
2.1	Fixed scope and primitive classifications	4
2.2	Derived substrate theorems	6
2.3	Elementary substrate facts	8
2.4	Structural closure of the fixed-substrate layer	9
2.5	Trajectory layer	10
3	Structural properties of coherent trajectories	12
4	Collapse and irrecoverability	15
5	Structural consequences of coherent construction	17
5.1	Bivalence and the exclusion of intermediate statuses	17
5.2	Tensor determination and skin neutrality	17
5.3	Endpoint certification and path memory	18
5.4	Composition factorization and non-aggregative behavior	19
5.5	Structural contrasts	19
6	Conclusion	20
A	Lean 4 correspondence and formalization provenance	21
A.1	Public surface	21
A.2	Primitive-theorem correspondence tables	22
A.3	Paper-facing wrappers	24
A.4	Interpretive remark on exactness	25
	References	26

1 Introduction

Mathematical treatments of admissible construction have often focused on static objects rather than histories: sets of sentences, consequence relations, deductive calculi, or probability assignments. Representative reference points include consequence-based logic, proof theory, subjective-probabilistic coherence, and theories of belief change and information dynamics [1–6, 9–12]. This paper does not present a theory of reasoning in the psychological or broad epistemic sense. It presents a trajectory theory of admissible construction on a fixed substrate. Construction unfolds as a sequence of construction states related by licensed transitions, identity-preserving transformations, and extensions. The central question is not whether a terminal state appears acceptable in isolation, but when a construction trajectory remains structurally coherent throughout its development.

Edition note. This scientific reference update retains the theorem sequence of the 20 April 2026 source edition. Its original companion titles [7,8] remain identified as historical references; the dated reference editions cited below identify the present foundational correspondence. Appendix A retains the formalization provenance documented in that source edition.

Dependency declaration. This paper works over a fixed substrate satisfying standing/admissibility equivalence, bivalence, identity-preserving tensor-preserving transformation, licensed same-scope transition, fail-closed enforcement, and irrecoverability under admissible continuation. These conditions are restated here because the trajectory theorems depend on them, but they are not derived afresh in this paper. Non-degenerate determinate construction necessarily incurs the kernel [13]; the complete interface and its continuation conditions are developed in [14]. The exact retained-dependency realization appears in the Kernel, Definition B.10, Theorem B.11, Lemma B.12, and Corollary B.13. The relation between historical retention and endpoint certification is stated in Structure, Theorem 7.2, and AMetric, Theorems 11.1–11.2 and Example 11.3 [15]. The present paper is the internal trajectory member of this dependency order: it derives the finite fixed-scope path structure once the recorded substrate is fixed. Its uniqueness claim concerns that substrate and the specified standing-preservation question.

Within that regime, our answer is bivalent and unexpectedly rigid. Coherence is identified with *standing preservation along legal construction trajectories*. The no-lower-generator theorem proved in Section 2 is therefore local to admissibility-preserving auxiliary classifiers on that same substrate; it is not a cross-regime foundational theorem. No new primitive is introduced at the trajectory level. Instead, coherence is defined as a derived property: a trajectory is coherent exactly when every state along it retains standing. What the later sections show is that, once this definition is coupled to the fixed substrate, both the positive and negative structure of coherent construction become sharply constrained.

For standalone readability, the paper records the fixed substrate directly rather than making the reader reconstruct it from external companion work. This is a convenience for local proof order, not a claim that the substrate first arises here. Section 2 fixes scope, states the standing/admissibility interface, introduces same-scope parameters and the anchor/tensor/skin role discipline, defines tensor content and reachability, records the governing fixed-substrate theorems for licensed transitions, and proves the local no-lower-generator theorem needed for the trajectory layer. The trajectory theory is then derived from that recorded substrate. Accordingly, every proposition, lemma, theorem, and corollary after Section 2 is proved internally from the stated definitions and the recorded fixed-substrate theorems. Appendix A records the archived Lean 4 correspondence at the level of public theorem names, theorem families, modules, and documented dependency spines.

The intended scope is deliberately narrow. Every definition and theorem is internal to one fixed substrate and concerns finite nonempty trajectories only. No claim is made here about cross-

scope transport, branching dynamics, infinite developments, probabilistic enrichment, or alternative regimes with different scope data.

Within that setting, the theory has a rigid shape. Coherent trajectories are closed under prefix extraction and legal composition, invariant under identity-preserving transformation, tensor-determined rather than skin-dependent, and irrecoverably lost upon any standing failure. In particular, once any prefix of a trajectory is incoherent, no admissible continuation can restore coherence. The resulting picture is not merely that trajectories matter; it is that, in the present regime, coherent construction has an exact finite path geometry rather than a loose family of acceptable developments.

This framework therefore differs sharply from static consequence accounts, probabilistic coherence programs, repair calculi, and update semantics that leave path status as an external appraisal [1–3,10,11]. In the present regime, coherence is a binary structural property of admissible construction paths, and no alternative path classification consistent with the fixed substrate yields a different coherence structure.

The paper proceeds as follows. Section 2 states the fixed-scope substrate and introduces trajectories, legality, and coherence. Section 3 develops the positive structure of the theory, including prefix preservation, composition closure, and invariance under identity-preserving tensor-preserving transformation. Section 4 develops the negative structure, culminating in the no-deferred-coherence theorem and the irrecoverability of coherence failure. Section 5 draws out the main consequences for theories of construction, including tensor determination, skin neutrality, endpoint certification, and the distinction between trajectory coherence and static, probabilistic, or outcome-based appraisal. Section 6 concludes, and Appendix A records the Lean 4 formalization interface used by the fixed substrate.

2 Formal setup

We now record the concrete fixed-scope substrate from which the trajectory theory is derived. Dependency declaration: the paper works throughout over a substrate satisfying standing/admissibility equivalence, bivalence, identity-preserving tensor-preserving transformation, licensed same-scope transition, fail-closed enforcement, and irrecoverability under admissible continuation. These conditions are stated here because every later trajectory theorem depends on them. They are not re-derived from a weaker base in this paper. Their foundational setting and exact continuation criteria are given in the Kernel and Structure reference editions [13,14]. The present section therefore serves two roles: it makes the local proof base explicit, and it isolates the strictly local closure theorem that closes the trajectory layer against internal admissibility-preserving reinterpretation.

2.1 Fixed scope and primitive classifications

We work at a fixed scope D . Here *scope* means the carrier specification under which admissibility is evaluated: the relevant domain, construction envelope, role commitments, and licensing conditions are held fixed throughout. A *same-scope move* is any move that purports to preserve that specification.

Definition 2.1 (Standing and admissibility). At fixed scope D , an *admissibility judgment* is a bivalent determination *admit/reject*. We write

$$A(x)$$

for the admissibility status of a state x at that scope, and

$$S(x)$$

for its standing status. *Standing-positive* means admitted; *standing-negative* means rejected.

Definition 2.2 (Same-scope parameter). Within a fixed admitted scope, a *same-scope parameter* is any selector, tolerance, deferred-enforcement knob, privileged representative, repair mechanism, or other admissibility-bearing degree of freedom whose variation purports to preserve scope while altering standing, legality, or consequence.

Definition 2.3 (Anchor / tensor / skin). A *role assignment* is a typing of interior content into:

- (i) *anchors*: non-negotiable preconditions whose violation collapses standing at the current scope;
- (ii) *tensors*: load-bearing compatibility relations that carry standing-relevant constraint content; and
- (iii) *skin*: post-admission descriptive surplus, including bookkeeping, presentation, and narrative, whose removal does not change tensor content.

Definition 2.4 (Tensor content and standing invariants). Fix an admitted scope with state-set X . For $x \in X$, the *tensor content* of x , written $\text{Ten}(x)$, is the total standing-relevant compatibility content carried by x at that scope: the conjunction of all anchor and tensor clauses satisfied by x , modulo skin. An *invariant* at the fixed scope is any predicate or value determined solely by $\text{Ten}(x)$.

Definition 2.5 (Standing equivalence). For $x, y \in X$, declare

$$x \sim y \iff \text{Ten}(x) = \text{Ten}(y).$$

Equivalently, x and y are *standing-equivalent* exactly when they carry the same tensor content.

Definition 2.6 (Identity-preserving transformation). An *identity-preserving transformation* is any post-admission map used only for interior bookkeeping or presentation. Such a transformation is admissible at the fixed scope if and only if it preserves tensor content, equivalently if it preserves standing equivalence in the sense of Definition 2.5.

Definition 2.7 (Licensed same-scope transition). A relation

$$\rightsquigarrow \subseteq X \times X$$

is called a *licensed same-scope transition relation* if $x \rightsquigarrow y$ means that y is reached from x by a same-scope construction move admitted by the fixed role and licensing discipline. In particular, such a move may use only licensed operator envelopes, may not introduce a same-scope parameter in the sense of Definition 2.2, may not depend on illicit identity-preserving transformation, and may not perform post hoc repair.

Remark 2.8 (Move license does not imply standing preservation). The relation $\rightsquigarrow$ classifies construction-step forms, not already-certified standing outcomes. We do *not* assume that

$$x \rightsquigarrow y \Rightarrow S(x) \quad \text{or} \quad x \rightsquigarrow y \Rightarrow S(y).$$

A licensed same-scope step may therefore terminate in a standing-negative state. This is what makes legal-but-incoherent trajectories available for study and prevents coherence from being built into legality.

Definition 2.9 (Finite same-scope reachability). For states $x, y \in X$, write

$$x \rightsquigarrow y$$

if and only if there exist $n \geq 0$ and states $x_0, \dots, x_n \in X$ such that

$$x = x_0, \quad y = x_n, \quad x_i \rightsquigarrow x_{i+1} \text{ for all } 0 \leq i < n.$$

Thus $\rightsquigarrow$ is the finite same-scope reachability relation generated by $\rightsquigarrow$.

Definition 2.10 (Primitive substrate). A *fixed-scope primitive substrate* is a tuple

$$\mathbf{P}_D = (X, A, S, \rightsquigarrow, \sim, \text{Ten})$$

consisting of:

- (i) a set X of construction states at the fixed scope,
- (ii) an admissibility predicate $A : X \rightarrow \mathbf{Prop}$,
- (iii) a standing predicate $S : X \rightarrow \mathbf{Prop}$,
- (iv) a licensed same-scope transition relation $\rightsquigarrow$ on X ,
- (v) a standing-equivalence relation $\sim$ on X , and
- (vi) a tensor-content assignment Ten inducing $\sim$.

The present paper works over a fixed primitive substrate $\mathbf{P}_D = (X, A, S, \rightsquigarrow, \sim, \text{Ten})$ of this form.

Remark 2.11 (Conventions and quantifier scope). Unless explicitly stated otherwise, every quantifier below ranges over the fixed state-set X , every appearance of $\rightsquigarrow$, $\sim$, and $\rightsquigarrow$ is interpreted relative to $\mathbf{P}_D$, and every trajectory is finite and nonempty. None of the results below compares distinct substrates or invokes cross-scope identifications.

2.2 Derived substrate theorems

The fixed substrate is governed by the following recorded substrate theorems. In the present paper, “recorded substrate theorem” means a theorem treated as part of the fixed substrate before the trajectory derivations begin; it does not mean that the theorem is first derived here or that every stronger governance-level closure question has already been settled locally. The trajectory layer does not re-derive these results from a weaker base. Appendix A records their archived Lean 4 correspondence, which is exact theorem-level in some cases and theorem-family or module-level in others.

Recorded Substrate Theorem 2.12 (Standing–admissibility equivalence). *For every $x \in X$,*

$$A(x) \leftrightarrow S(x).$$

Thus admissibility and standing coincide extensionally at the fixed scope, even when they are kept notationally distinct for explanatory purposes.

Recorded Substrate Theorem 2.13 (Bivalence and fail-closed enforcement). *For every $x \in X$,*

$$S(x) \vee \neg S(x).$$

Moreover, under-specified, unlicensed, or ambiguous moves are rejected rather than repaired.

Recorded Substrate Theorem 2.14 (No generators, no carriers, no repairs). *Standing cannot be generated, transported across scopes by association, or recovered inside a rejected lineage. Any move that requires post hoc repair is inadmissible.*

Recorded Substrate Theorem 2.15 (AMetric boundary). *At the boundary only coherent classification and standing determination occur. No metric notions, operators, rates, or analytic mechanisms exist at the boundary as standing-bearing objects.*

Recorded Substrate Theorem 2.16 (Licensed interior bookkeeping). *Metric descriptors and analytic operators may appear inside an admitted scope only as licensed bookkeeping. Such bookkeeping does not itself generate standing-bearing force.*

Recorded Substrate Theorem 2.17 (Operators as envelopes). *Operators exist only as licensed envelopes with specified domain, invariants, failure conditions, and revocation conditions. Unlicensed operator use is inadmissible.*

Recorded Substrate Theorem 2.18 (Identity-preserving transformation neutrality). *Redescription is post-admission only. A redescription step is admissible only if it is lawful in the sense of Definition 2.6, that is, only if it preserves tensor content at the fixed scope. In particular,*

$$x \sim y \Rightarrow (S(x) \leftrightarrow S(y)).$$

No partial identity-preserving transformation and no gain of constructional power under such transformation are permitted.

Recorded Substrate Theorem 2.19 (Composition discipline). *There is no retroactivity and no partial composition. If an intermediate state fails standing, the lineage cannot be repaired downstream into coherence.*

Recorded Substrate Theorem 2.20 (Deterministic enforcement). *Violations of the preceding recorded substrate theorems are structurally detectable. Enforcement is fail-closed and admits no in-lineage repair.*

Recorded Substrate Theorem 2.21 (Structural exhaustion). *Once the scope is fixed, no new admissibility level, atomic bundle, admissible question, or operator family may be introduced.*

Recorded Substrate Theorem 2.22 (Irrecoverability under admissible continuation). *For all $x, y \in X$,*

$$(x \rightsquigarrow y) \wedge \neg S(x) \Rightarrow \neg S(y).$$

Recorded Substrate Theorem 2.22 is negative absorption for the stated continuation relation. Retaining a failed original stage already prevents the retained history from becoming wholly coherent; certifying that history from its terminal standing additionally uses this absorption property. Structure, Theorem 7.2, and AMetric, Theorems 11.1–11.2, establish the exact distinction [14, 15]. AMetric, Example 11.3, realizes absorption when states are complete retained histories and transitions append legal stages. The Kernel’s required-dependency construction gives a second explicit realization: complete predecessor lists and sound local checks determine support, and a failed required ancestor remains a failed conjunct at every descendant retaining it [13, Definition B.10, Theorem B.11, Lemma B.12, and Corollary B.13].

For an already licensed complete governing use, faithful ATS bookkeeping retains its original application conditions, required dependencies, and warrants. It inherits their proved preservation without a second qualification event; the relevant realization is the preserved complete use, not the

role names alone [16, Theorems 3.3, 3.5, 6.3, and Corollary 6.5]. A fresh corrected construction may carry its own standing while the original failed occurrence remains failed. It does not thereby confer positive standing on the old retained dependency record. The absorption property concerns the retained governing lineage under its declared continuation relation.

Remark 2.23 (Primitive economy and local closure). The fixed substrate is intentionally small and, in the limited sense formalized by Theorem 2.30 below, closed against stricter admissibility-preserving standing classifiers on the same fixed substrate. Its standing commitments are fixed scope, standing and admissibility, bivalence, fail-closed enforcement, the AMetric boundary, licensed operator existence, identity-preserving transformation neutrality, composition discipline, deterministic enforcement, structural exhaustion, and irrecoverability. The theorem below does *not* claim that every possible construction regime must begin with exactly this package. It says that once the present scope data and recorded substrate theorems are fixed, any admissibility-preserving auxiliary classifier must remain constant on tensor-equivalence classes and therefore cannot serve as a deeper standing-bearing layer within this regime. In that precise local sense, no further admissibility-preserving refinement of standing exists within this substrate. That local closure is what makes the trajectory layer mathematically closed under the only admissibility-preserving reinterpretations available inside the regime. The trajectory theory proved later is therefore not an informal overlay on the substrate but the internally forced path theory of that fixed base.

Remark 2.24 (Why the trajectory problem is posed on this substrate). The objects studied below are same-scope legal construction histories whose coherence is standing preservation and whose failures are irreversible under admissible continuation. A system lacking fixed-scope standing/admissibility classification, identity-preserving transformation, licensed same-scope transition, fail-closed enforcement, or irrecoverability does not support that class of trajectories *as those notions are used in this paper*; it supports a different process theory, not a counterexample to the present one. In that sense the present substrate is not an arbitrary restriction on a wider trajectory question. It is the regime needed for the specific coherence problem studied here to be well-posed. The Kernel establishes necessary governance and supplies the retained required-dependency realization; Structure gives the exact interface and transition conditions [13, 14]. The present paper takes the stated substrate as fixed input.

2.3 Elementary substrate facts

The primitive definitions above already entail a few structural properties that will be used implicitly later. We record them once here so that every later derivation rests on explicit in-paper preliminaries.

Proposition 2.25 (Standing equivalence is an equivalence relation). *The relation $\sim$ of Definition 2.5 is an equivalence relation on X .*

Proof. By Definition 2.5, $x \sim y$ holds exactly when $\text{Ten}(x) = \text{Ten}(y)$. Reflexivity, symmetry, and transitivity therefore follow from the corresponding properties of equality. $\square$

Proposition 2.26 (Finite reachability is reflexive and transitive). *The relation $\rightsquigarrow$ of Definition 2.9 is reflexive and transitive on X .*

Proof. Reflexivity is immediate by taking $n = 0$ in Definition 2.9. For transitivity, suppose $x \rightsquigarrow y$ and $y \rightsquigarrow z$. Choose witnessing finite chains from x to y and from y to z . Concatenating those chains at the shared state y yields a finite licensed same-scope chain from x to z , so $x \rightsquigarrow z$. $\square$

Corollary 2.27 (Tensor-determined invariants are constant on $\sim$ -classes). *If I is an invariant in the sense of Definition 2.4 and $x \sim y$, then $I(x) = I(y)$.*

Proof. By definition, an invariant is determined solely by tensor content. If $x \sim y$, then $\text{Ten}(x) = \text{Ten}(y)$, so every tensor-determined quantity takes the same value at x and y . $\square$

2.4 Structural closure of the fixed-substrate layer

A persistent hostile reading would treat the fixed substrate above as merely the point at which the paper chooses to stop. To block that reading in the representation class relevant to the present trajectory theory, we record a closure theorem showing that no stricter admissibility-preserving standing layer exists beneath the present tensor content while the fixed-scope regime is held unchanged.

Definition 2.28 (Admissibility-preserving auxiliary representation). Fix the substrate $\mathbf{P}_D$. An *admissibility-preserving auxiliary representation* of standing on $\mathbf{P}_D$ consists of a set Y , a map $I : X \rightarrow Y$, and a predicate $\Phi : Y \rightarrow \mathbf{Prop}$ such that

$$S(x) \leftrightarrow \Phi(I(x)) \quad \text{for every } x \in X,$$

while the fixed-scope data $X, A, S, \rightsquigarrow, \sim, \text{Ten}$ and Recorded Substrate Theorems 2.12–2.22 are left unchanged.

Definition 2.29 (Strictly lower standing generator). An admissibility-preserving auxiliary representation (Y, I, Φ) is called a *strictly lower standing generator* if its standing-bearing discrimination is strictly finer than tensor content, that is, if there exist $x, y \in X$ such that

$$\text{Ten}(x) = \text{Ten}(y) \quad \text{but} \quad \Phi(I(x)) \not\leftrightarrow \Phi(I(y)).$$

Equivalently, the standing-bearing partition induced by I refines the tensor partition strictly.

Theorem 2.30 (No further refinement of standing within the fixed substrate / local structural non-derivability). *Within the fixed substrate $\mathbf{P}_D$, there exists no strictly lower standing generator. Equivalently, no further admissibility-preserving refinement of standing exists there beyond tensor content. More precisely, if (Y, I, Φ) is any admissibility-preserving auxiliary representation of standing on $\mathbf{P}_D$, then for all $x, y \in X$,*

$$\text{Ten}(x) = \text{Ten}(y) \Rightarrow (\Phi(I(x)) \leftrightarrow \Phi(I(y))).$$

Consequently there is a well-defined predicate

$$\widehat{\Phi} : \text{Ten}(X) \rightarrow \mathbf{Prop}$$

such that for every $x \in X$,

$$S(x) \leftrightarrow \widehat{\Phi}(\text{Ten}(x)).$$

In particular, any admissibility-preserving attempt to derive standing from an auxiliary invariant collapses to tensor-determined bookkeeping and yields no deeper standing-bearing layer within this substrate.

Proof. Let (Y, I, Φ) be an admissibility-preserving auxiliary representation, and suppose that $\text{Ten}(x) = \text{Ten}(y)$. By Definition 2.5, this means $x \sim y$. Recorded Substrate Theorem 2.18 therefore gives

$$S(x) \leftrightarrow S(y).$$

On the other hand, Definition 2.28 yields

$$S(x) \leftrightarrow \Phi(I(x)) \quad \text{and} \quad S(y) \leftrightarrow \Phi(I(y)).$$

Combining these equivalences gives

$$\Phi(I(x)) \leftrightarrow \Phi(I(y)).$$

This proves the first claim and rules out the possibility of a strictly lower standing generator in the sense of Definition 2.29.

Now define $\hat{\Phi}$ on the image $\text{Ten}(X)$ by choosing any $x \in X$ with $\text{Ten}(x) = t$ and setting

$$\hat{\Phi}(t) :\Longleftrightarrow \Phi(I(x)).$$

The first part shows that this is well-defined: if $\text{Ten}(x) = \text{Ten}(y) = t$, then $\Phi(I(x)) \leftrightarrow \Phi(I(y))$. For any $x \in X$, taking $t = \text{Ten}(x)$ yields

$$\hat{\Phi}(\text{Ten}(x)) \leftrightarrow \Phi(I(x)) \leftrightarrow S(x).$$

Thus standing factors through tensor content. The final sentence follows: an auxiliary invariant may at most repackage tensor-determined classification, not supply a deeper standing-bearing generator. $\square$

The factorization just proved concerns the decoded predicate $\Phi \circ I$, not the entire auxiliary map I . Different auxiliary records may therefore retain different metadata while giving the same standing answer on a tensor class. No minimality of I is asserted. For a complete family of original uses, ATS distinguishes complete representations from minimal ones and permits faithful surplus [16, Theorems 3.3 and 3.5]; the present theorem concerns the fixed standing predicate specified in Definition 2.28.

Remark 2.31 (Scope of the no-lower-generator theorem). Theorem 2.30 says exactly this: no further refinement of standing exists within the present fixed substrate, in the sense of admissibility-preserving auxiliary representations. That is the exact local closure statement needed for the trajectory layer: once this result is in place, the later trajectory theorems do not remain vulnerable to a stricter standing-bearing reinterpretation drawn from within the same regime. It does not by itself settle broader governance-level questions about whether the substrate itself is derivable from below, and it does not analyze cross-scope, history-sensitive, or substrate-changing enrichments that depart from Definition 2.28.

Remark 2.32 (Closure versus extra metadata). Theorem 2.30 does not forbid attaching additional metadata to states. It forbids only a stricter *standing-bearing* layer beneath tensor content. Any auxiliary data that leaves the standing-bearing partition unchanged is, within the fixed-scope regime, bookkeeping or skin rather than a deeper admissibility level.

2.5 Trajectory layer

We now pass from the primitive substrate to the objects studied in the rest of the paper.

Definition 2.33 (Trajectory). A *trajectory* is a finite nonempty sequence

$$\tau = (x_0, x_1, \dots, x_n)$$

with each $x_i \in X$. If $\tau = (x_0, \dots, x_n)$, we write $|\tau| = n$ for its terminal index and $\tau_i = x_i$ for its i -th state.

Definition 2.34 (Legal trajectory). A trajectory $\tau = (x_0, \dots, x_n)$ is *legal* if and only if

$$\forall i < n, x_i \rightsquigarrow x_{i+1}.$$

We write $\text{Legal}(\tau)$.

Remark 2.35 (Legality versus coherence). Legality and coherence are distinct. A legal trajectory is one whose steps are same-scope and licensed in the transition sense of Definition 2.7. A coherent trajectory is a legal trajectory whose states preserve standing throughout. By Recorded Substrate Theorem 2.12, a pointwise definition of legality requiring standing at every state would collapse coherence into legality and trivialize the trajectory theory.

Remark 2.36 (Why legal trajectories may still collapse). Remarks 2.8 and 2.35 should be read together. The relation $\rightsquigarrow$ marks a same-scope step as belonging to the licensed construction envelope; it does not guarantee preservation of standing. Theorems about incoherence therefore concern trajectories whose *form of continuation* remains licensed even though standing has failed at some state in the lineage.

Definition 2.37 (Trajectory realization of reachability). For states $x, y \in X$, the primitive reachability relation $x \rightsquigarrow y$ of Definition 2.9 holds if and only if there exists a legal trajectory $\tau = (x_0, \dots, x_n)$ such that $x_0 = x$ and $x_n = y$.

Definition 2.38 (Prefix). Let $\tau = (x_0, \dots, x_n)$ and $\tau' = (y_0, \dots, y_k)$. We say that τ' is a *prefix* of τ , and write $\text{Prefix}(\tau', \tau)$, if and only if $k \leq n$ and

$$y_i = x_i \quad \text{for all } 0 \leq i \leq k.$$

Definition 2.39 (Extension). We say that σ is an *extension* of τ , and write $\text{Extends}(\sigma, \tau)$, if and only if $\text{Prefix}(\tau, \sigma)$.

Definition 2.40 (Composition). Let $\tau_1 = (x_0, \dots, x_m)$ and $\tau_2 = (y_0, \dots, y_n)$. We say that τ_1 and τ_2 are *composable* if and only if

$$x_m = y_0.$$

In that case their *composition* is the trajectory

$$\tau_1 \circ \tau_2 := (x_0, \dots, x_m, y_1, \dots, y_n).$$

When $n = 0$, this convention yields $\tau_1 \circ \tau_2 = \tau_1$.

Definition 2.41 (Trajectory identity-preserving equivalence). Let $\tau = (x_0, \dots, x_n)$ and $\tau' = (y_0, \dots, y_m)$. We say that τ and τ' are *pointwise identity-preserving equivalent*, and write

$$\tau \approx \tau',$$

if and only if $n = m$ and

$$x_i \sim y_i \quad \text{for all } 0 \leq i \leq n.$$

Definition 2.42 (Coherent trajectory). A legal trajectory $\tau = (x_0, \dots, x_n)$ is *coherent* if and only if every state on the trajectory has standing:

$$\text{Coherent}(\tau) :\iff \text{Legal}(\tau) \wedge \forall i \leq |\tau|, S(\tau_i).$$

This is the central derived definition of the paper.

Definition 2.43 (Failure witness). A *failure witness* for a trajectory τ is an index $i \leq |\tau|$ such that

$$\neg S(\tau_i).$$

A legal trajectory is therefore coherent exactly when it contains no failure witness.

Only Recorded Substrate Theorems 2.13, 2.18, and 2.22 are used directly in the trajectory proofs that follow. Theorem 2.30 plays a different role: it closes the present fixed-substrate layer against stricter admissibility-preserving standing reinterpretations on the same fixed substrate. The remaining substrate theorems still matter: they specify the intended regime over which the trajectory theory is claimed and rule out readings in which metric bookkeeping, cross-scope transport, or post hoc repair are allowed to bear standing-level force.

No further fixed-substrate assumptions are introduced after this section. Every proposition, lemma, theorem, and corollary in the remainder of the paper is derived from Definitions 2.1–2.43 and Recorded Substrate Theorems 2.12–2.22. Given this substrate—recorded here for local use, with its retained-dependency realization and exact transition conditions identified in [13, 14]—the remainder of the paper derives the trajectory theory forced internally. It is therefore conditional and internal: it analyzes what follows once this fixed substrate is in place, not what follows in every alternative regime. Within that scope, however, the resulting trajectory layer is intended to be complete at the level of finite fixed-scope path structure addressed here. Appendix A records the documented Lean 4 correspondence for this fixed substrate layer.

The definitions above already show the sense in which the framework is bivalent. Coherence is not introduced as a scalar, metric, or graded notion. It is a predicate on legal construction paths: a legal trajectory either preserves standing throughout or it does not.

3 Structural properties of coherent trajectories

We now establish the positive structural facts about coherence over the fixed substrate of Section 2. Throughout, legality means legality in the sense of Definition 2.34: each adjacent step is a licensed same-scope transition, and coherence means legality together with standing preservation at the fixed scope. All claims in this section are internal to $\mathbf{P}_D$ and concern finite trajectories only.

Proposition 3.1 (Coherence characterization). *Let τ be a legal trajectory. Then*

$$\text{Coherent}(\tau) \iff \forall i \leq |\tau|, S(\tau_i).$$

Proof. Since τ is assumed legal, the legality clause in Definition 2.42 is already satisfied. The definition therefore reduces to the condition that every state on τ has standing at the fixed scope. $\square$

Proposition 3.2 (Failure-witness characterization). *Let τ be a legal trajectory. Then*

$$\neg \text{Coherent}(\tau) \iff \exists i \leq |\tau|, \neg S(\tau_i).$$

Proof. By Proposition 3.1,

$$\neg \text{Coherent}(\tau) \iff \neg \forall i \leq |\tau|, S(\tau_i).$$

Because the index set $\{0, \dots, |\tau|\}$ is finite, this is equivalent to

$$\exists i \leq |\tau|, \neg S(\tau_i).$$

That index is precisely a failure witness in the sense of Definition 2.43. $\square$

In particular, every legal fixed-scope trajectory is either coherent or contains a failure witness. No intermediate standing-bearing status occurs.

Lemma 3.3 (Prefix legality). *If $\text{Legal}(\tau)$ and $\text{Prefix}(\tau', \tau)$, then $\text{Legal}(\tau')$.*

Proof. Write

$$\tau = (x_0, \dots, x_n), \quad \tau' = (x_0, \dots, x_k)$$

with $k \leq n$. Since τ is legal, each adjacent transition

$$x_i \rightsquigarrow x_{i+1}$$

with $0 \leq i < n$ already occurs in τ as a licensed same-scope step. In particular, for each $i < k$ the same transition holds inside τ' . Hence τ' satisfies Definition 2.34. $\square$

We may now pass from legality to coherence.

Theorem 3.4 (Prefix preservation). *If $\text{Coherent}(\tau)$ and $\text{Prefix}(\tau', \tau)$, then $\text{Coherent}(\tau')$.*

Proof. Since $\text{Coherent}(\tau)$, the trajectory τ is legal and every state on τ has standing. By Lemma 3.3, every prefix τ' of τ is legal. Since each state of τ' is also a state of τ , every state of τ' has standing as well. Therefore τ' is coherent by Definition 2.42. $\square$

Proposition 3.5 (Composition legality). *If $\text{Legal}(\tau_1)$, $\text{Legal}(\tau_2)$, and τ_1, τ_2 are composable, then $\text{Legal}(\tau_1 \circ \tau_2)$.*

Proof. Write

$$\tau_1 = (x_0, \dots, x_m), \quad \tau_2 = (y_0, \dots, y_n),$$

and assume $x_m = y_0$. Then

$$\tau_1 \circ \tau_2 = (x_0, \dots, x_m, y_1, \dots, y_n).$$

It remains to check the adjacent transitions. Within the initial block $x_0, \dots, x_m$, legality follows from $\text{Legal}(\tau_1)$. Within the final block $y_1, \dots, y_n$, legality follows from $\text{Legal}(\tau_2)$. At the interface, if $n \geq 1$, we need

$$x_m \rightsquigarrow y_1.$$

But $x_m = y_0$, and $\text{Legal}(\tau_2)$ gives $y_0 \rightsquigarrow y_1$. Hence the interface transition is licensed as well. Therefore $\tau_1 \circ \tau_2$ is legal. $\square$

Theorem 3.6 (Composition closure). *If $\text{Coherent}(\tau_1)$, $\text{Coherent}(\tau_2)$, and τ_1, τ_2 are composable, then*

$$\text{Coherent}(\tau_1 \circ \tau_2).$$

Proof. Since $\text{Coherent}(\tau_1)$ and $\text{Coherent}(\tau_2)$, both trajectories are legal and every state in each has standing. By Proposition 3.5, the composite trajectory $\tau_1 \circ \tau_2$ is legal.

Now every state of $\tau_1 \circ \tau_2$ is a state of either τ_1 or τ_2 . Since all states in τ_1 and τ_2 have standing, every state in $\tau_1 \circ \tau_2$ has standing. Hence $\tau_1 \circ \tau_2$ is coherent by Definition 2.42. $\square$

We next show that coherence is invariant under identity-preserving transformation. Because identity-preserving transformation is tensor-preserving by Definition 2.6, this is the point at which the anchor/tensor/skin discipline enters the trajectory theory directly.

Both trajectories in the following theorem are assumed legal. Tensor equality is used to preserve their standing judgments, not to infer a previously unlicensed transition or joint application. Where legality is inherited from a complete licensed envelope, its original domain and compatibility warrants remain in force under the faithful bookkeeping; the corresponding ATS descent conditions preserve both applicability and output [16, Theorem 6.3 and Corollary 6.5].

Theorem 3.7 (Identity-preserving transformation invariance). *Let τ and τ' be legal trajectories. If $\tau \approx \tau'$, then*

$$\text{Coherent}(\tau) \iff \text{Coherent}(\tau').$$

Proof. Write

$$\tau = (x_0, \dots, x_n), \quad \tau' = (y_0, \dots, y_n),$$

so that $x_i \sim y_i$ for each $0 \leq i \leq n$.

Assume first that $\text{Coherent}(\tau)$. Since τ is legal, Proposition 3.1 yields

$$S(x_i) \quad \text{for all } 0 \leq i \leq n.$$

By Recorded Substrate Theorem 2.18, each relation $x_i \sim y_i$ implies

$$S(x_i) \leftrightarrow S(y_i).$$

Hence $S(y_i)$ holds for all i . Since τ' is legal by hypothesis, Proposition 3.1 yields $\text{Coherent}(\tau')$.

The converse implication is symmetric: if $\text{Coherent}(\tau')$, then $S(y_i)$ for all i , and Recorded Substrate Theorem 2.18 again gives $S(x_i)$ for all i , whence $\text{Coherent}(\tau)$. Therefore

$$\text{Coherent}(\tau) \iff \text{Coherent}(\tau').$$

□

Corollary 3.8 (Tensor-determined coherence). *Let $\tau = (x_0, \dots, x_n)$ and $\tau' = (y_0, \dots, y_n)$ be legal trajectories. If*

$$\text{Ten}(x_i) = \text{Ten}(y_i) \quad \text{for all } 0 \leq i \leq n,$$

then

$$\text{Coherent}(\tau) \iff \text{Coherent}(\tau').$$

In particular, skin-only variation does not affect coherence.

Proof. By Definition 2.5, the hypothesis $\text{Ten}(x_i) = \text{Ten}(y_i)$ implies $x_i \sim y_i$ for each i . Hence $\tau \approx \tau'$ by Definition 2.41. The result now follows from Theorem 3.7. The final sentence is the special case in which the variation from x_i to y_i is purely skin-level and therefore leaves tensor content unchanged by Definition 2.3. □

Corollary 3.9 (Closure of coherent construction trajectories). *Let*

$$\mathcal{T}_{\text{coh}} := \{\tau : \text{Coherent}(\tau)\}.$$

Then $\mathcal{T}_{\text{coh}}$ is closed under prefix extraction and legal composition.

Proof. Closure under prefix extraction is Theorem 3.4. Closure under legal composition is Theorem 3.6. □

The results of this section establish the positive side of the theory. Coherence is stable under initial-segment extraction, under admissible gluing of same-scope trajectories, and under identity-preserving tensor-preserving transformation. In particular, coherence is not a fragile annotation on paths but a stable invariant under the admissible structural operations available within the regime.

Remark 3.10 (Limits of the closure results). Corollary 3.9 is an internal closure statement. It does *not* authorize gluing across distinct substrates, and it does not address branching or infinite constructions. Every appearance of prefix, composition, and identity-preserving transformation here remains relative to $\mathbf{P}_D$. No alternative path classification consistent with $\mathbf{P}_D$ yields a different coherence structure.

The next section develops the negative side: how coherence fails, how failure propagates, and why it cannot be recovered once lost.

4 Collapse and irrecoverability

Section 3 established the positive closure behavior of coherence. We now turn to the negative structure. Incoherence is witness-based and persistent: once a standing-failing state enters a legal same-scope trajectory, every trajectory that retains that history retains the failure as well. This section makes that claim precise and links it to Recorded Substrate Theorem 2.22.

Theorem 4.1 (Local failure implies trajectory incoherence). *Let τ be a legal trajectory. If τ contains a failure witness, then*

$$\neg \text{Coherent}(\tau).$$

Proof. By Definition 2.43, to say that τ contains a failure witness is to say that there exists an index $i \leq |\tau|$ such that $\neg S(\tau_i)$. Since τ is legal, Proposition 3.2 yields $\neg \text{Coherent}(\tau)$. $\square$

Theorem 4.2 (One-step standing loss implies incoherence). *Let $\tau = (x_0, \dots, x_n)$ be a legal trajectory. If for some $i < n$ one has*

$$S(x_i) \wedge \neg S(x_{i+1}),$$

then

$$\neg \text{Coherent}(\tau).$$

Proof. The hypothesis supplies a failure witness at index $i + 1$. The conclusion follows from Theorem 4.1. $\square$

The next lemma isolates the combinatorial fact behind no deferred coherence: extensions do not delete earlier states.

Lemma 4.3 (Failure witnesses persist under extension). *Let τ and σ be trajectories such that $\text{Extends}(\sigma, \tau)$. If τ contains a failure witness, then σ contains a failure witness.*

Proof. Write $\tau = (x_0, \dots, x_n)$. Since $\text{Extends}(\sigma, \tau)$ means $\text{Prefix}(\tau, \sigma)$ by Definition 2.39, there exists a representation

$$\sigma = (x_0, \dots, x_n, z_1, \dots, z_m)$$

for some $m \geq 0$. If $i \leq n$ satisfies $\neg S(x_i)$, then the same index i witnesses failure in σ because $\sigma_i = x_i$. $\square$

Theorem 4.4 (No deferred coherence). *Let τ be a legal trajectory and let τ' be a prefix of τ . If*

$$\neg \text{Coherent}(\tau'),$$

then

$$\neg \text{Coherent}(\tau).$$

Proof. By Lemma 3.3, every prefix of a legal trajectory is legal; hence τ' is legal. Proposition 3.2 then yields an index $i \leq |\tau'|$ with $\neg S(\tau'_i)$. Since τ' is a prefix of τ , the trajectory τ extends τ' in the sense of Definition 2.39. By Lemma 4.3, the same failure witness occurs in τ . Because τ is legal, Proposition 3.2 now gives $\neg \text{Coherent}(\tau)$. $\square$

The preceding theorem says that later continuation cannot improve an already incoherent prefix. This is the negative core of the trajectory theory. The next theorem sharpens that statement into a general extension principle.

Theorem 4.5 (Irrecoverability of coherence failure). *Let τ and σ be legal trajectories. If*

$$\text{Extends}(\sigma, \tau) \quad \text{and} \quad \neg \text{Coherent}(\tau),$$

then

$$\neg \text{Coherent}(\sigma).$$

Proof. Because τ is legal and incoherent, Proposition 3.2 yields a failure witness on τ . By Lemma 4.3, that witness persists in every extension σ of τ . Since σ is legal, Proposition 3.2 implies $\neg \text{Coherent}(\sigma)$. $\square$

Remark 4.6 (Internal extension principle). Theorem 4.5 concerns extensions that remain inside the same fixed substrate and preserve the earlier lineage as a prefix. It is not a theorem about cross-scope reinterpretation or about replacing the original trajectory by a different one. The claim is narrower and stronger: within a retained same-scope lineage, later continuation does not erase earlier failure.

The next result shows that the primitive irrecoverability theorem also yields a stronger state-level propagation theorem inside any legal trajectory.

Theorem 4.7 (Downstream propagation of standing failure). *Let $\tau = (x_0, \dots, x_n)$ be a legal trajectory. If i and j satisfy*

$$0 \leq i \leq j \leq n \quad \text{and} \quad \neg S(x_i),$$

then

$$\neg S(x_j).$$

Proof. Consider the finite sequence

$$\rho := (x_i, x_{i+1}, \dots, x_j).$$

Because τ is legal, every adjacent transition in ρ is legal as well. Hence ρ is a legal trajectory from x_i to x_j . By Definition 2.37, this means $x_i \rightsquigarrow x_j$. Recorded Substrate Theorem 2.22 therefore yields $\neg S(x_j)$ from $\neg S(x_i)$. $\square$

Corollary 4.8 (Earliest failure controls the tail). *Let $\tau = (x_0, \dots, x_n)$ be a legal trajectory such that $\neg \text{Coherent}(\tau)$. Then there exists a least index $i_* \leq n$ such that $\neg S(x_{i_*})$, and for every j with $i_* \leq j \leq n$ one has*

$$\neg S(x_j).$$

Proof. Since τ is legal and incoherent, Proposition 3.2 shows that the set

$$W := \{i \in \{0, \dots, n\} : \neg S(x_i)\}$$

is nonempty. Because W is a nonempty finite set of natural numbers, it has a least element i_* . The final claim follows from Theorem 4.7: for every $j \geq i_*$ with $j \leq n$, the failure of standing at x_{i_*} propagates to x_j . $\square$

In a legal fixed-scope trajectory, standing failure marks a genuine collapse point. Any extension preserves the failure witness, and the fixed-substrate irrecoverability principle forces that failure to propagate through all downstream states reachable from it. In this sense coherence is fail-closed: once lost, it cannot be regained by same-scope continuation.

5 Structural consequences of coherent construction

The preceding sections show that coherence is a binary, fail-closed property of legal construction paths. This section records the main structural consequences within the finite fixed-scope setting already specified.

5.1 Bivalence and the exclusion of intermediate statuses

Theorem 5.1 (Bivalence of coherence). *For every legal trajectory τ ,*

$$\text{Coherent}(\tau) \vee \neg\text{Coherent}(\tau).$$

Proof. Let $\tau = (x_0, \dots, x_n)$ be legal. By Recorded Substrate Theorem 2.13, each standing claim $S(x_i)$ is bivalent. Since the index set $\{0, \dots, n\}$ is finite, either all of the formulas $S(x_i)$ hold or at least one of them fails. In the first case, Proposition 3.1 yields $\text{Coherent}(\tau)$. In the second case, Proposition 3.2 yields $\neg\text{Coherent}(\tau)$. $\square$

Corollary 5.2 (No partial coherence). *For legal trajectories, coherence is not a primitive graded or intermediate status. Every legal trajectory is either coherent or incoherent.*

Proof. This is an immediate restatement of Theorem 5.1. $\square$

The force of Corollary 5.2 is not merely logical. It excludes any attempt to treat coherence itself as a primitive scalar quantity. One may define secondary summaries such as defect counts or collapse depth, but within the present framework such quantities are descriptive overlays on a prior binary fact.

5.2 Tensor determination and skin neutrality

Because identity-preserving transformation preserves tensor content, coherence is determined by tensor content and blind to skin. This is one of the paper’s strongest structural consequences: once tensor content is fixed, coherence is fixed with it throughout the present regime.

Corollary 5.3 (No coherence by skin repair). *Let τ and τ' be legal trajectories such that $\tau \approx \tau'$. If*

$$\neg\text{Coherent}(\tau),$$

then

$$\neg\text{Coherent}(\tau').$$

Proof. Suppose toward contradiction that $\text{Coherent}(\tau')$. Then Theorem 3.7 yields $\text{Coherent}(\tau)$, contradicting the hypothesis. Hence $\neg\text{Coherent}(\tau')$. $\square$

Remark 5.4 (Skin-blindness of coherence). Corollary 5.3 and Corollary 3.8 formalize a strict restriction on the theory. Once tensor content is fixed, altering only skin—presentation, bookkeeping, notation, narrative, or other post-admission descriptive surplus in the sense of Definition 2.3—cannot change the coherence status of a legal trajectory. In particular, incoherence cannot be repaired by restatement alone.

Remark 5.5 (AMetric consequence). Axioms 2.15 and 2.16 sharpen Corollary 5.2. A numerical trajectory score may be introduced as internal bookkeeping inside an already admitted scope, but it does not function as a primitive standing-bearing determinant of coherence. Boundary classification remains binary.

5.3 Endpoint certification and path memory

Although coherence is defined over whole trajectories, the irrecoverability structure of Section 4 yields a strong endpoint criterion. The point of the next theorem is not to retreat to static evaluation, but to show that in the present fail-closed regime terminal standing compresses an admissible history whose failures cannot be hidden or repaired downstream. The criterion is history-sensitive rather than static: terminal standing certifies the trajectory only because, in the present fail-closed regime, standing at the terminal state already excludes any earlier unrepaired collapse along that same lineage.

Theorem 4.7 applies Recorded Substrate Theorem 2.22; the converse direction of the terminal criterion below then uses that propagation. This is the negative-absorption case of Structure, Theorem 7.2, and AMetric, Theorem 11.2 [14,15]. For retained required-dependency paths, Kernel Corollary B.13 supplies the same endpoint certification from the support recurrence [13]. A projection to the latest independent stage may discard the retained history and need not satisfy that criterion.

Theorem 5.6 (Terminal-state criterion). *Let $\tau = (x_0, \dots, x_n)$ be a legal trajectory. Then*

$$\text{Coherent}(\tau) \iff S(x_n).$$

Proof. If $\text{Coherent}(\tau)$, then every state on τ has standing by Proposition 3.1, so in particular $S(x_n)$ holds.

Conversely, assume $S(x_n)$. Suppose toward contradiction that τ is not coherent. Since τ is legal, Proposition 3.2 yields an index $i \leq n$ such that $\neg S(x_i)$. Theorem 4.7 then implies $\neg S(x_n)$, contradicting the hypothesis. Hence τ must be coherent. $\square$

Corollary 5.7 (Shared endpoints force shared coherence). *Let $\tau = (x_0, \dots, x_m)$ and $\sigma = (y_0, \dots, y_n)$ be legal trajectories in the same fixed substrate with the same terminal state:*

$$x_m = y_n.$$

Then

$$\text{Coherent}(\tau) \iff \text{Coherent}(\sigma).$$

Proof. By Theorem 5.6,

$$\text{Coherent}(\tau) \iff S(x_m) \quad \text{and} \quad \text{Coherent}(\sigma) \iff S(y_n).$$

Since $x_m = y_n$, the two standing conditions are identical. The result follows. $\square$

Remark 5.8 (Literal endpoint identity matters). Corollary 5.7 uses literal identity of terminal states inside a single substrate. It does not assert that semantically similar endpoints drawn from different scopes or different state spaces must share coherence status. The theorem is strict because the paper is strict about substrate identity.

Corollary 5.9 (Stagewise terminal criterion). *Let $\tau = (x_0, \dots, x_n)$ be a legal trajectory. Then for every k with $0 \leq k \leq n$,*

$$\text{Coherent}((x_0, \dots, x_k)) \iff S(x_k).$$

Proof. For each $k \leq n$, the initial segment $(x_0, \dots, x_k)$ is a prefix of τ . By Lemma 3.3, it is therefore legal. The claim now follows by applying Theorem 5.6 to that prefix. $\square$

Remark 5.10 (Why endpoint certification does not collapse the theory). Theorem 5.6 does not reduce coherence to ordinary endpoint evaluation in the usual static sense, and it should not be read as a theorem about arbitrary path theories. The certifying endpoint predicate is not mere consistency, utility, agreement, or any other external terminal feature. It is standing, and standing is constrained here by admissibility, tensor content, and irrecoverability on the fixed substrate. The theorem is compressive only because the regime is history-sensitive: once failure is irrecoverable, terminal standing already carries the effect of the admissible history leading to that endpoint. In that sense the endpoint criterion is not a weakening of the trajectory theory but one of its sharpest signatures.

5.4 Composition factorization and non-aggregative behavior

The closure theorem of Section 3 can be sharpened. When a legal composite is coherent, coherence does not arise at the level of the whole from incoherent parts; it factorizes exactly through the components.

Theorem 5.11 (Exact factorization under composition). *Let $\tau_1 = (x_0, \dots, x_m)$ and $\tau_2 = (y_0, \dots, y_n)$ be legal and composable. Then*

$$\text{Coherent}(\tau_1 \circ \tau_2) \iff \text{Coherent}(\tau_1) \wedge \text{Coherent}(\tau_2).$$

Proof. Assume first that $\text{Coherent}(\tau_1 \circ \tau_2)$. Since τ_1 is a prefix of $\tau_1 \circ \tau_2$, Theorem 3.4 gives $\text{Coherent}(\tau_1)$.

It remains to prove $\text{Coherent}(\tau_2)$. Because τ_2 is legal by hypothesis, it is enough by Proposition 3.1 to show that every state on τ_2 has standing. The interface state y_0 equals x_m , which appears in $\tau_1 \circ \tau_2$ and therefore has standing. Every state y_i with $1 \leq i \leq n$ also appears in $\tau_1 \circ \tau_2$ and therefore has standing. Hence every state on τ_2 has standing, so τ_2 is coherent.

Conversely, if both τ_1 and τ_2 are coherent, then Theorem 3.6 yields $\text{Coherent}(\tau_1 \circ \tau_2)$. $\square$

Corollary 5.12 (No coherence by aggregation). *Let τ_1 and τ_2 be legal and composable. If either τ_1 or τ_2 is incoherent, then*

$$\neg \text{Coherent}(\tau_1 \circ \tau_2).$$

Proof. Suppose toward contradiction that $\text{Coherent}(\tau_1 \circ \tau_2)$. Then Theorem 5.11 implies that both τ_1 and τ_2 are coherent, contrary to hypothesis. $\square$

Corollary 5.12 shows that coherence is not an emergent averaging property of concatenation. Appending a coherent suffix to an incoherent prefix does not repair the whole, and prefixing a coherent segment to an incoherent suffix does not domesticate the latter.

5.5 Structural contrasts

Theorems 5.1, 5.6, 5.11, and 4.5 sharply constrain what a trajectory theory of construction can look like within the present fixed-substrate setting.

First, the theory is not graded. A construction episode is not assigned a primitive degree of coherence. There may be earlier or later collapse points, shorter or longer coherent prefixes, and more or fewer downstream states affected by failure, but these are structural features of a trajectory whose coherence status is already settled.

Second, the theory is not retrospective. By Theorem 4.5, once a legal trajectory becomes incoherent, no extension can restore coherence. Later explanatory success, agreement, convenience,

or other favorable suffix properties may attach to an extension, but they do not erase the inherited failure witness.

Third, the theory is not aggregative. Theorem 5.11 shows that coherence of a composite trajectory factorizes through coherence of its components. A coherent whole cannot be assembled out of an incoherent part by adding enough well-behaved surrounding material.

Fourth, the theory does not coincide with familiar static endpoint appraisals. Section 4 shows that standing failure propagates through legal continuation, while Theorem 5.6 shows that terminal standing certifies the full trajectory only because standing is history-sensitive. The endpoint does not license the path by appearing acceptable in isolation; it certifies the path only when the primitive standing predicate has already excluded concealed collapse along the way.

The object of evaluation is a finite construction path. Endpoint certification is available only because the primitive standing predicate retains the memory of admissible failure. Coherence is therefore neither a soft summary of the whole path nor a property recoverable from arbitrary terminal virtues. The combined force of bivalence, factorization, and irrecoverability is that coherent construction acquires a determinate finite path structure inside the regime rather than a flexible family of approximately acceptable histories. In particular, the present theory diverges from static consequence accounts, probabilistic coherence programs, revision-theoretic repair models, and dynamic-information logics precisely where those traditions let closure conditions, numerical regularity, revision postulates, or update operators do work that is done here by standing preservation on a fixed substrate [1–3, 5, 6, 10–12]. The result is a binary structural invariant of legal constructional development, tensor-determined and skin-blind at fixed scope. These are claims about the stated substrate, not omnibus claims about every possible formal treatment of construction.

6 Conclusion

This paper has developed a bivalent trajectory theory of coherent construction over an explicit fixed-scope substrate. It is the internal, downstream member of a layered package. The Kernel establishes necessary governance and retained required-dependency support [13]; Structure and AMetric identify the exact interface and endpoint conditions [14, 15]. The present paper does not redo that work. Its claim is narrower and mathematical, but also more exact: given the substrate recorded in Section 2, it isolates the finite fixed-scope path structure of coherence itself. Within that regime, coherence is defined not as a new primitive, metric, or retrospective appraisal, but as standing preservation along a legal construction trajectory. From that definition, together with the standing/admissibility interface, identity-preserving transformation, fail-closed enforcement, and irrecoverability, both the positive and negative structure of coherent construction follow.

On the positive side, coherent trajectories are stable under prefix extraction, legal composition, and identity-preserving tensor-preserving transformation. On the negative side, incoherence is witness-based, no deferred coherence is possible, and coherence failure is irrecoverable under legal extension. Once a failure witness appears, the trajectory has already ceased to be coherent, and no admissible continuation changes that fact. Taken together, these results show that coherence in the present setting is not merely defined along paths; it is rigidly organized by the path structure itself.

Within that fixed-substrate setting, the framework treats construction in process terms rather than merely state terms. Even where terminal certification is available, it is available only because standing is already constrained by admissible history. A terminal state does not vindicate an arbitrary path by appearing satisfactory in isolation; it certifies the path only when the underlying standing predicate has already excluded hidden collapse along the way. The endpoint criterion is therefore not a retreat from trajectory analysis but the strongest compression result the regime

permits.

The claims are intentionally narrow. The paper proves results for finite nonempty trajectories internal to one fixed substrate. It does not attempt to settle cross-scope comparison, branching construction, interacting agents, infinite developments, or what alternative substrates with different scope data might support. Those are natural next questions, but they are not smuggled into the present conclusions.

Within that restricted setting, the resulting theory is rigid. Coherent construction is not a matter of degree, later endorsement, successful aggregation, or skin-level reformulation. It is a binary structural property of legal constructional development, determined by tensor content at fixed scope. Theorem 2.30 establishes the local closure claim needed for the trajectory layer: no further admissibility-preserving refinement of standing exists within the same fixed substrate beyond tensor content. In that sense the paper delivers, for the regime it studies, a complete finite fixed-scope trajectory theory rather than a loose collection of path lemmas. The present paper should therefore be read as follows: the substrate is defined here, justified elsewhere, and the trajectory layer is derived here. Broader governance-level closure claims and broader compositional forcing claims are distinct from, and stronger than, the local fixed-substrate theorem proved in this paper.

A Lean 4 correspondence and formalization provenance

This appendix retains the correspondence documented in the 20 April 2026 source edition between the paper’s fixed-scope substrate and the companion Lean 4 development. Its role is documentary rather than derivational: none of the mathematical arguments in the body of the paper depends on the appendix, but the appendix makes explicit which theorem names, files, and dependency layers were documented as witnessing the fixed substrate inside the formalization. Theorem 2.30 proves, within the representation class used in this paper, that any admissibility-preserving auxiliary representation of standing collapses to tensor-determined bookkeeping. The appendix should therefore be read as provenance documentation for the locally closed substrate used by the trajectory theory, not as evidence that the formalization has merely paused at an arbitrary intermediate layer and not as a substitute for broader governance-level closure arguments.

Two cautions are important. First, the paper intentionally uses a compressed fixed-scope notation. The Lean development is richer: some paper-level fixed-substrate theorems correspond to a *family* of Lean results rather than to a single theorem with the same name. Second, the table below records the exact names documented in the archived handoff packets and public theorem register. Where the paper-level fixed-substrate is a semantic compression over several Lean files, the correspondence is stated at theorem-family level rather than by inventing a nonexistent one-to-one theorem identifier.

The tables and exported names below preserve that historical provenance. The dated reference editions cited in the body supply the present mathematical correspondence; this reference update makes no claim that the archived Lean branch has been rebuilt or that those newer manuscripts have been matched to its exports.

A.1 Public surface

The public surface documented in that edition is:

```
MaleyLean.lean, importing Foundation, TheoremRegister, and PaperStatements.
```

The archived theorem register re-exports canonical public names from the capstone files `AmetricCollapseTheorem`, `Bivalence`, and `BivalenceUniqueness`, while `PaperStatements.lean`

provides paper-facing wrappers of the form `Paper_...`

The documented dependency spine of the archived branch is:

```
Core → Boundary → AmetricBoundary → Transport → Witness
→ NoRepair → Irrecoverability → StandingDeterminacy → Bivalence → BivalenceUniqueness.
```

with the additional metric-separation branch

```
AmetricBoundary + StandingDeterminacy
→ StandingMetricSeparation → NoInformationInFailureRegion
→ StandingCarriesMetricFreedom → AmetricCollapseTheorem
→ TheoremRegister → PaperStatements.
```

A.2 Primitive-theorem correspondence tables

The archived correspondences divide naturally into two classes. Table 1 records paper-level fixed-substrates whose Lean provenance is documented at exact public theorem name or clearly identified theorem-family level. Table 2 then records the remaining fixed-scope primitives whose present Lean provenance is module-level, architectural, or otherwise many-to-one rather than a single public export. Splitting the documentation this way keeps the exact public surface visible without forcing theorem-level and infrastructure-level correspondences into one oversized table.

Table A.1. Public theorem and theorem-family correspondences

Table 1: Public theorem-level and theorem-family correspondences.

Paper fixed-substrate theorem	Documented Lean theorem(s)	File or module	Dependency spine / correspondence note
Bivalence and fail-closed enforcement	<code>StandingBivalenceTheorem</code> ; internal support by <code>standing_bivalence</code> , <code>standing_failure_bivalence</code> , <code>no_third_classification_status</code> .	<code>TheoremRegister.lean</code> ; <code>Bivalence.lean</code> ; <code>BivalenceUniqueness.lean</code>	<code>StandingDeterminacy → Bivalence → BivalenceUniqueness → TheoremRegister → PaperStatements</code> . The fail-closed reading is the branch design principle attached to this bivalent partition.
No generators, no carriers, no repairs	<code>failure_witness_blocks_legality</code> , <code>no_metric_repair_at_witnessed_failure</code> , <code>failure_witness_cannot_be_repaired_by_metrics</code> ; strengthened in <code>irrecoverable_failure_blocks_metric_repair</code> .	<code>NoRepair.lean</code> ; <code>Irrecoverability.lean</code>	<code>Witness → NoRepair → Irrecoverability</code> . The paper’s “no generators / no carriers / no repairs” clause is a compressed reading of this theorem family together with the no-transport design discipline recorded in the handoff.

continued on next page

Paper fixed-substrate theorem	Documented Lean theorem(s)	File or module	Dependency spine / correspondence note
AMetric boundary	<code>AmetricCollapseTheorem</code> , <code>StandingRegionCarriesAllAdmissibleMetricContent</code> , <code>FailureRegionSupportsNoAdmissibleMetricContent</code> ; internal support by <code>ametric_collapse_theorem</code> .	<code>AmetricCollapseTheorem.lean</code> ; public export via <code>TheoremRegister.lean</code>	<code>AmetricBoundary</code> + <code>StandingDeterminacy</code> $\rightarrow$ <code>StandingMetricSeparation</code> $\rightarrow$ <code>NoInformationInFailureRegion</code> $\rightarrow$ <code>StandingCarriesMetricFreedom</code> $\rightarrow$ <code>AmetricCollapseTheorem</code> $\rightarrow$ <code>TheoremRegister</code> $\rightarrow$ <code>PaperStatements</code> .
Licensed interior bookkeeping	<code>StandingRegionCarriesAllAdmissibleMetricContent</code> , <code>FailureRegionSupportsNoAdmissibleMetricContent</code> .	<code>TheoremRegister.lean</code> ; derived from <code>StandingMetricSeparation</code> , <code>NoInformationInFailureRegion</code> , <code>StandingCarriesMetricFreedom</code> .	These capstones formalize the paper’s claim that metric content is licensed only inside the standing region and becomes trivial on failure regions.
Redescription neutrality	<code>failure_witness_blocks_legality</code> , <code>silent_redescription_failure_is_irrecoverable</code> , <code>irrecoverable_failure_blocks_legality</code> .	<code>NoRepair.lean</code> ; <code>NoSilentRedescription.lean</code> ; <code>Irrecoverability.lean</code>	<code>Transport</code> / <code>Witness</code> $\rightarrow$ <code>NoRepair</code> $\rightarrow$ <code>NoSilentRedescription</code> $\rightarrow$ <code>Irrecoverability</code> . At paper level this is summarized as identity-preserving tensor-preserving transformation and standing invariance under such transformation.
Structural exhaustion	<code>BivalenceUniquenessTheorem</code> ; internal support by <code>standing_classification_is_unique</code> , <code>standing_failure_partition_is_unique</code> , <code>no_admissible_third_status</code> .	<code>BivalenceUniqueness.lean</code> ; public export via <code>TheoremRegister.lean</code>	<code>Bivalence</code> $\rightarrow$ <code>BivalenceUniqueness</code> $\rightarrow$ <code>TheoremRegister</code> $\rightarrow$ <code>PaperStatements</code> . The paper’s stronger fixed-scope exhaustion statement reads this uniqueness family together with the AMetric and no-parameterization discipline.
Irrecoverability under admissible continuation	<code>IrrecoverabilityIsMetricInvariant</code> ; internal support by <code>irrecoverable_failure_iff_failure_witness</code> , <code>irrecoverable_failure_blocks_legality</code> , <code>irrecoverable_failure_blocks_metric_repair</code> , <code>irrecoverable_failure_forces_metric_collapse</code> , <code>irrecoverability_is_metric_invariant</code> .	<code>Irrecoverability.lean</code> ; public export via <code>TheoremRegister.lean</code>	<code>Witness</code> $\rightarrow$ <code>NoRepair</code> $\rightarrow$ <code>Irrecoverability</code> $\rightarrow$ <code>TheoremRegister</code> $\rightarrow$ <code>PaperStatements</code> . The paper rewrites this theorem family in fixed-scope trajectory language.

Table A.2. Structural and module-level correspondences

Table 2: Structural, semantic-compression, and module-level correspondences.

Paper fixed-substrate theorem	Documented Lean theorem(s)	File or module	Dependency spine / correspondence note
Standing–admissibility equivalence	No exact one-line theorem name is exposed in the archived public handoff; the branch documentation instead treats standing as the operative public classifier and summarizes the system as implementing “Admissibility (standing).”	Foundation / StandingDeterminacy (documented at module level)	This paper-level fixed-substrate is therefore recorded as a semantic identification built into the archived public reading of the branch rather than as a separately named public theorem.
Operators as envelopes	No single public theorem name is documented in the retrieved register; the operative content sits at infrastructure level in the legality and transport stack.	Foundation / Boundary / Transport (module level)	The paper-level theorem compresses the formal legality infrastructure governing identity-preserving transformations and operator envelopes. A one-name public export is not documented in the archived handoff packet.
Composition discipline	The archived handoff identifies Composition as a foundation module and names ClosurePropagation.lean as the next propagation-facing public extension; no single public theorem name for the paper’s full composition discipline is documented in the retrieved packet.	Composition (foundation module); planned ClosurePropagation.lean extension	The paper-level fixed-substrate compresses the already-formalized composition infrastructure plus the irrecoverability / no-repair stack. The absence of a documented public theorem name here is a property of the archived handoff packet, not of the paper’s mathematics.
Deterministic enforcement	No single public theorem name is exposed; documented in the branch design principles as fail-closed architecture, no silent retargeting, and no repair.	Project-wide design layer; reflected in NoRepair / Irrecoverability family	This primitive is represented formally by the combination of bivalence, legality blocking, and no-repair theorems rather than by one isolated export.

A.3 Paper-facing wrappers

The archived branch also documents a paper-facing export layer in `PaperStatements.lean`. Among the exact names documented there are:

- `Paper_AmetricCollapseTheorem`
- `Paper_StandingRegionCarriesAllAdmissibleMetricContent`
- `Paper_FailureRegionSupportsNoAdmissibleMetricContent`
- `Paper_IrrecoverabilityIsMetricInvariant`

- `Paper_StandingBivalenceTheorem`
- `Paper_BivalenceUniquenessTheorem`

Those wrappers matter because they show that the formal branch already distinguishes between internal theorem engineering and paper-facing theorem presentation.

A.4 Interpretive remark on exactness

The appendix should be read with one final distinction in view. The body of the paper is written in fixed-scope trajectory language because that is the most economical mathematical surface for the present article. The Lean development, by contrast, is organized around a richer semantic substrate involving systems, identity-preserving transformations, witnesses, legality, `AMetric` boundaries, metric separation, and irrecoverability. Accordingly, “exact correspondence” should be read in the sense appropriate to formal extraction: where the paper states a theorem that is literally one of the public exports, the match is theorem-to-theorem; where the paper compresses several lower-level Lean facts into a single mathematical primitive, the correspondence is exact at theorem-family and dependency level rather than at single-name level.

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